

TELECOM DATABASE CASE STUDY

Scenario

A telecom company manages:

- Subscribers (customers)
- SIM cards
- Call/Data usage
- Billing

◆ PART 1: DDL (Data Definition Language)

? Q1. Create database telecom_db

? Q2. Create table subscribers

```
CREATE TABLE subscribers (  
  subscriber_id INT AUTO_INCREMENT PRIMARY KEY,  
  subscriber_name VARCHAR(100) NOT NULL,  
  phone_number VARCHAR(20) UNIQUE,  
  city VARCHAR(50),  
  registration_date DATE  
);
```

? Q3. Create table plans

```
CREATE TABLE plans (  
  plan_id INT AUTO_INCREMENT PRIMARY KEY,  
  plan_name VARCHAR(100),  
  monthly_fee DECIMAL(10,2),  
  data_limit INT  
);
```

? Q4. Create table usage_records

```
CREATE TABLE usage_records (  
  usage_id INT AUTO_INCREMENT PRIMARY KEY,  
  subscriber_id INT,  
  data_used DECIMAL(10,2),
```

```
call_minutes INT,  
usage_date DATE,  
FOREIGN KEY (subscriber_id) REFERENCES subscribers(subscriber_id)  
);
```

? Q5. Add column email to subscribers

```
ALTER TABLE subscribers  
ADD email VARCHAR(100);
```

? Q6. Modify column city

```
ALTER TABLE subscribers  
MODIFY city VARCHAR(100);
```

? Q7. Drop column email

```
ALTER TABLE subscribers  
DROP COLUMN email;
```

◆ PART 2: DML (Data Manipulation Language)

? Q8. Insert subscribers

```
INSERT INTO subscribers (subscriber_name, phone_number, city, registration_date)  
VALUES  
( 'Aung Aung', '0912345678', 'Yangon', '2025-01-01'),  
( 'Su Su', '0923456789', 'Mandalay', '2025-02-01'),  
( 'Kyaw Kyaw', '0934567890', 'Yangon', '2025-03-01');
```

? Q9. Insert plans

```
INSERT INTO plans (plan_name, monthly_fee, data_limit)  
VALUES  
( 'Basic', 10, 5),  
( 'Standard', 20, 10),  
( 'Premium', 30, 20);
```

? Q10. Insert usage records

```
INSERT INTO usage_records (subscriber_id, data_used, call_minutes, usage_date)  
VALUES  
(1, 2.5, 30, '2025-04-01'),  
(1, 1.0, 10, '2025-04-02'),
```

```
(2, 5.0, 60, '2025-04-01'),  
(3, 8.0, 90, '2025-04-03');
```

? Q11. Update data usage

```
UPDATE usage_records  
SET data_used = data_used + 1  
WHERE subscriber_id = 1;
```

? Q12. Delete inactive subscriber

```
DELETE FROM subscribers  
WHERE subscriber_id = 3;
```

◆ PART 3: DQL (Data Query Language)

? Q13. Select all subscribers

```
SELECT * FROM subscribers;
```

? Q14. Find subscribers in Yangon

```
SELECT * FROM subscribers  
WHERE city = 'Yangon';
```

? Q15. Total data usage per subscriber

```
SELECT subscriber_id, SUM(data_used) AS total_data  
FROM usage_records  
GROUP BY subscriber_id;
```

? Q16. Subscribers with high usage (>5 GB)

```
SELECT subscriber_id, SUM(data_used) AS total_data  
FROM usage_records  
GROUP BY subscriber_id  
HAVING SUM(data_used) > 5;
```

? Q17. Sort subscribers by registration date

```
SELECT * FROM subscribers
ORDER BY registration_date DESC;
```

? Q18. Join subscriber with usage

```
SELECT s.subscriber_name, u.data_used, u.call_minutes
FROM subscribers s
JOIN usage_records u
ON s.subscriber_id = u.subscriber_id;
```

◆ PART 4: TCL (Transaction Control Language)

? Q19. Data correction with rollback

```
START TRANSACTION;
```

```
UPDATE usage_records
SET data_used = data_used + 10
WHERE subscriber_id = 2;
```

```
ROLLBACK;
```

? Q20. Confirm update with commit

```
START TRANSACTION;
```

```
UPDATE usage_records
SET data_used = data_used + 2
WHERE subscriber_id = 2;
```

```
COMMIT;
```

? Q21. Use SAVEPOINT

```
START TRANSACTION;
```

```
UPDATE usage_records SET data_used = data_used + 1 WHERE subscriber_id=1;
SAVEPOINT sp1;
```

```
UPDATE usage_records SET data_used = data_used + 5 WHERE subscriber_id=1;
```

```
ROLLBACK TO sp1;
```

COMMIT;

◆ **PART 5: DCL (Data Control Language)**

? **Q22. Create user**

```
CREATE USER 'telecom_user'@'localhost' IDENTIFIED BY 'password123';
```

? **Q23. Grant SELECT access**

```
GRANT SELECT ON telecom_db.* TO 'telecom_user'@'localhost';
```

? **Q24. Grant INSERT and UPDATE**

```
GRANT INSERT, UPDATE ON telecom_db.usage_records  
TO 'telecom_user'@'localhost';
```

? **Q25. Revoke UPDATE permission**

```
REVOKE UPDATE ON telecom_db.usage_records  
FROM 'telecom_user'@'localhost';
```

? **Q26. Show permissions**

```
SHOW GRANTS FOR 'telecom_user'@'localhost';
```

? **Q27. Drop user**

```
DROP USER 'telecom_user'@'localhost';
```